

CHKDSK Loop Fix

Issue: CHKDSK runs after every reboot. The volume is flagged dirty and won't clear.

Key correction: do NOT use `chkdsk C: /f /r /x` on the live system drive. The `/x` switch forces a dismount, which the running OS cannot do to C:, so the check is deferred to boot time - and if it fails to clear the dirty bit (very common with Fast Startup on), it re-runs every boot. That is usually the CAUSE of the loop, not the fix. Use `/x` only on non-system volumes that can actually dismount.

Decision Tree

```
CHKDSK runs every reboot
■ ■ ■ Check dirty bit: fsutil dirty query C:
■ ■ ■ "Volume - C: is Dirty"
■ ■ ■ Is Fast Startup ON? (most common cause of the loop)
■ ■ ■ YES → turn it OFF (Fix A step 1), reboot, recheck dirty bit
■ ■ ■ NO → run online scan: chkdsk C: /scan
■ ■ ■ Cleared → Resolved. Monitor.
■ ■ ■ Still dirty → Fix B (offline repair in WinRE)
■ ■ ■ "Volume - C: is NOT Dirty" but CHKDSK still runs
■ ■ ■ Boot-time schedule is stuck → Fix C (clear BootExecute / chkntfs)
```

FIX A: Break the loop on the live system drive (do this first)

Step 1 - turn off Fast Startup (it does a hybrid shutdown that skips the full dirty-bit reset, so the check never clears):

CMD:

```
powercfg /h off
reg add "HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Power" /v HiberbootEnabled /t REG_DWORD /d 0
```

PowerShell:

```
powercfg /h off
Set-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Control\Session Manager\Power" -Name HiberbootEnabled -Value 0
```

Step 2 - run an ONLINE scan (no dismount, does not schedule a boot check).

This is the correct command for the system drive:

CMD / PowerShell (same):

```
chkdsk C: /scan
```

`/scan` runs against the live volume and can clear soft dirty-bit conditions. If it reports it fixed problems, reboot and re-check `fsutil dirty query C:.`

Step 3 - if the bit is now clear, restore default boot behavior:

```
chkntfs /d
```

Recheck:

```
fsutil dirty query C:
```

"NOT Dirty" → resolved.

FIX B: Offline repair (when /scan cannot clear it)

The system drive can only be fully repaired while it is offline, in the Recovery Environment (WinRE), where C: is genuinely dismounted.

1. Boot to WinRE: Settings > System > Recovery > Advanced startup > Restart now, OR hold Shift while clicking Restart, OR boot Windows install media.
2. Troubleshoot > Advanced options > Command Prompt.
3. In WinRE the system drive may be lettered D: - confirm first:

```
bcdedit | find "osdevice"
```

4. Run the full repair against the correct letter (only here is **/f** **/r** on the system drive appropriate, because it is offline):

```
chkdsk C: /f /r
```

Let it complete all five stages (can take 1-4 hours on a large or failing disk).

5. Reboot normally and verify:

```
fsutil dirty query C:
```

*/x is still not needed here - WinRE already has the volume offline. Reserve **/x** for non-system data volumes (e.g. **chkdsk D: /f /x**) that can dismount live.*

FIX C: Dirty bit is clear but CHKDSK still runs at boot

This means a stale boot-time schedule, not a dirty volume.

Check what is scheduled:

```
chkntfs C:
```

- "...will be checked at next reboot" or "...is dirty" → a check is queued.

Exclude the drive from the boot check (stops the scheduled run):

```
chkntfs /x C:
```

Reboot. If CHKDSK no longer runs, the schedule was the problem.

Restore default behavior afterward (so real dirty volumes still get checked):

```
chkntfs /d
```

If it persists, inspect the BootExecute key directly. The default value is exactly

`autocheck autochk *`. Extra entries force checks:

CMD:

```
reg query "HKLM\SYSTEM\CurrentControlSet\Control\Session Manager" /v BootExecute
```

PowerShell:

```
Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Control\Session Manager" -Name BootExecute | Select-Object
```

If it shows anything other than `autocheck autochk *` (for example `autocheck autochk /r \??\C:`),

reset it to default:

CMD:

```
reg add "HKLM\SYSTEM\CurrentControlSet\Control\Session Manager" /v BootExecute /t REG_MULTI_SZ /d "autocheck autochk *
```

PowerShell:

```
Set-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Control\Session Manager" -Name BootExecute -Value "autocheck autochk *
```

Reboot.

FIX D: Dirty bit keeps coming BACK after clearing

Something is re-dirtying the volume - usually a failing disk or a filter driver.

1. Check for NTFS corruption events:

```
wevtutil qe System /q:"*[System[Provider[@Name='Ntfs'] and (EventID=55 or EventID=137)]" /c:10 /f:txt
```

2. Check physical drive health:

PowerShell:

```
Get-PhysicalDisk | Select-Object FriendlyName, MediaType, HealthStatus, OperationalStatus
```

CMD (WMIC is deprecated but present on many builds):

```
wmic diskdrive get Model,Status
```

If HealthStatus is "Warning"/"Unhealthy" or Status is not "OK" → back up immediately

and replace the drive. A recurring dirty bit on healthy-testing hardware points at a

third-party disk/AV filter driver - remove or update it.

Prevention

- Keep **Fast Startup OFF** on machines that have shown the loop; the hybrid shutdown is the single most common reason the dirty bit never clears.

- Always shut down cleanly - abrupt power loss sets the dirty bit.
- Recurring loops on the same machine usually mean a degrading drive; replace proactively.

Windows Endpoint Remediation Toolkit — Author: Adrian Melendez

Windows Endpoint Remediation Toolkit